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This paper has developed a simple model of the spread of misinformation over social media platforms. A group of Bayesian agents with heterogeneous priors receive and share news items (articles) according to a stochastic sharing network, determined by the social media platform. Articles may be truthful and informative about an underlying state, or may contain misinformation, making them (weakly) anti-correlated with the underlying state. Upon receiving an article, an agent can decide to share it with others, ignore it, or actively call out another agent for propagating misinformation (“dislike”). Misinformation spreads when agents share articles expecting positive social media feedback and little negative reactions.

Though simple and parsimonious, the model encapsulates several rich strategic interactions. Agents receive utility from sharing truthful articles and not misinformation, but also enjoy peer engagement with shared content. The former force pushes agents toward sharing pro-attitudinal content, whereas the latter force pushes agents toward pleasing those in their sharing network. The ideological congruence between an agent and those in her sharing network, which we capture with the notion of homophily, is critical for sharing decisions. Because individuals are more likely to dissent against articles that disagree with their prior beliefs, an agent will be more cautious in sharing articles that disagree with the views of those in her sharing network.

We provide several comparative static results. Some of those are very intuitive, though still useful for interpreting a range of results in the emerging empirical literature on social media and misinformation. For example, we find that while misinformation typically spreads less than truthful content (holding all else constant), more sensational content tends to be shared more. Moreover, when misinformation is correlated with sensationalism, the rapid spread misinformation can be problematic.

Of particular interest are the comparative statics with respect to homophily—measuring whether agents tend to share articles with others with similar or dissimilar beliefs. We show that when there is a highly-reliable article, an increase in homophily reduces the virality of content. Because this article is unlikely to contain misinformation, it is of broad appeal to a wide range of social media users, independent of ideology. An increase in homophily then reduces the extent to which this article can spread throughout the sharing network. The implications of homophily for low-reliability articles are very different, however: in a well-connected network, such articles will be disliked and stopped by users who disagree with their message, and anticipating this behavior and the loss of reputation they can suffer from spreading misinformation, even those who agree with their message would not share them widely. In contrast, high homophily creates echo chambers, where users share low-reliability messages aligned with their beliefs, because they understand that there are few negative reputational consequences from doing so. Misinformation contained in low-reliability articles can then spread virally in these echo chambers.

Our framework enables a tractable study of platform incentives in designing algorithms that determine who shares with whom. To do this, we assume that the platform aims to maximize user engagement (which is a good approximation to the objectives of major social media platforms such as Facebook or Twitter). Our main result is a striking one. When an article is highly reliable, the

platform chooses a sharing network with minimal homophily to maximize the spread and appeal of the content throughout the user community. In contrast to this case, when the relevant articles have lower reliability, the platform chooses a network with maximal homophily and recommends articles to users with aligned beliefs. These articles then spread rapidly in the “filter bubble” the platform’s algorithms have created—because now ideologically like-minded individuals know that they are unlikely to be caught sharing misinformation in their extreme echo chambers.

We also study regulations aimed at minimizing the spread of misinformation. Revealing the provenance of a news item (for example, providing full context for a quote) can be useful, because this additional information allows users to more easily fact-check the content for veracity. However, we also show that this type of regulation may backfire because of a “bystander effect” whereby future users assume content has been previously fact-checked and put more faith in it. Similarly, tagging or censoring articles from less reliable sources may make agents more confident in those that remain on the site untagged—thus creating a Bayesian version of the “false sense of security”. Finally, we show that regulating platform algorithms that shape sharing network properties (in particular, homophily) can be effective.

Our framework was purposefully chosen to be simple and several generalizations would be interesting to consider in future work. Most importantly, our assumption that agents are Bayesian rational should be viewed as a useful, albeit not fully realistic, benchmark. In our setting, it brought out certain new strategic forces—highlighting how social media actions exhibit strategic complementarity and how the degree of homophily alters agents’ strategic behavior. Nevertheless, misinformation may be particularly damaging when agents have cognitive limitations and are only boundedly rational. Incorporating such considerations is one of the most important directions for future research. Interesting questions that emerge in this case relate to whether the platform, in addition to designing algorithms that create filter bubbles, may also choose strategies that exploit the cognitive limitations of users.

Other theoretical generalizations that might be interesting to consider include extensions to repeated interactions with incomplete information, which would enable agents to also update their beliefs about the ideological position of other agents in their sharing network. Fully endogenizing reputational concerns, taking into account the network position of agents, would be another interesting direction for future research. In this case, the existing reputational capital of an agent will determine how likely she is to risk sharing misinformation. We can also use this extended setup with repeated interactions to study how agents update their initial political views. When there is limited misinformation, agents will gradually learn the true state. In contrast, when there is a significant probability of misinformation, agents will be uncertain about how to interpret articles that disagree with their priors and this may place an upper bound on the speed and possibility of learning (see [Acemoglu et al. \(2016\)](#)).

Despite its simplicity, our model makes several new empirical predictions, most notably related to the non-monotonic effects of homophily and polarization and to platform incentives and algorithmic decisions. Investigating these predictions empirically as well as generating new stylized facts about patterns of these information cascades on social media, is another important area for future research.